

Speak Like A Mathematician:

distance from a point to a line:

perpendicular bisector:

equidistant:

Theorems About Perpendicular Lines

Linear Pair Perpendicular Theorem: if two lines intersect to form a linear pair of _____ angles, then the lines are _____.

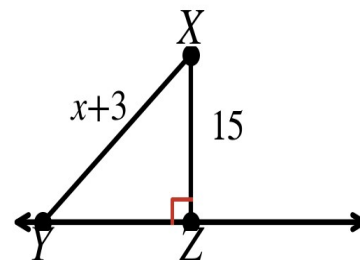
Perpendicular Transversal Theorem: in a plane, if a transversal is perpendicular to one of two _____ lines, then it is perpendicular to the _____.

Lines Perpendicular to a Transversal Theorem: in a plane, if two lines are perpendicular to the _____ line, then they are _____ to each other.

Example 1 - Distance from a Point to a Line

In the diagram, $X\bar{Z} \perp Y\bar{Z}$, $XY = x + 3$, and $XZ = 15$.

- Name the shortest segment from point X to line YZ.
- Write and solve an inequality for x.
- Explain why XY can never equal 15.

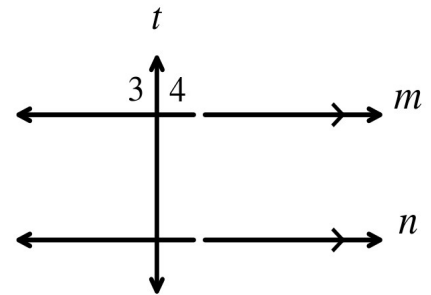


Example 2 - Using the Perpendicular Transversal Theorem

Given: $m \parallel n$, $\angle 3 \cong \angle 4$

Prove: $t \perp n$

Statements	Reasons
1. $\angle 3 \cong \angle 4$	1. Given
2. $\angle 3$ and $\angle 4$ form a linear pair	2.
3.	3.
4.	4.
5.	5.

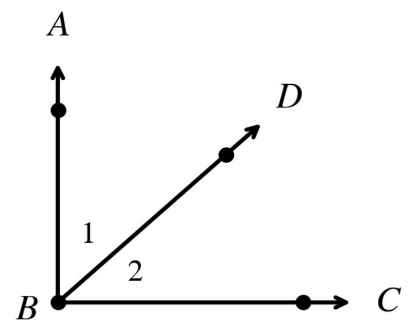


Example 3 - A Proof with Angle Addition

Given: $\overline{BA} \perp \overline{BC}$, $m\angle 1 = m\angle 2$

Prove: $m\angle 1 = 45^\circ$

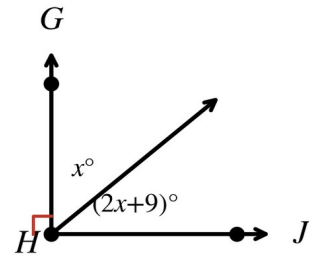
Statements	Reasons
1. $\overline{BA} \perp \overline{BC}$	1. Given
2.	2.
3.	3.
4.	4.
5.	5.
6.	6.
7.	7.
8.	8.



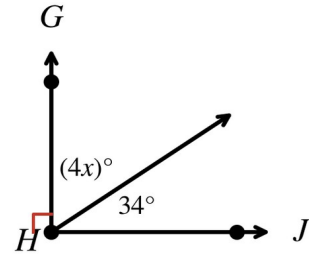
Example 4 - Find the Value of x

In each diagram, $\overline{GH} \perp \overline{HJ}$. Find the value of x. Show your work.

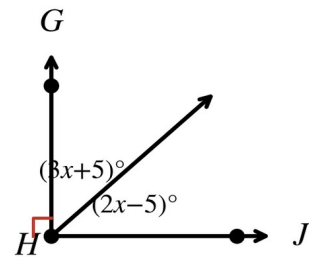
a)



b)



c)



Example 5 - Reasoning with the Theorems

Answer each question and name the theorem that supports your answer.

- In a plane, line $p \perp$ line r and line $q \perp$ line r . What can you conclude about p and q ?
- In a plane, line $a \parallel$ line b and line $c \perp$ line a . What can you conclude about c and b ?
- Two lines intersect to form a linear pair of congruent angles. Find the measure of each angle.